

PHYTOINTELLIGENCE 3.5

Standalone Multivariate R&D Framework & Forensic Pharmacology Specification

System Version	3.5.0 (Full Standalone Release)	Release Date	June 10, 2026
Architecture Owner	Marie-Soleil Seshat Landry	Epistemic Standard	1=1=1 Axiom / Seshat's Axiom
Protocol Status	LHFP-II Compliant (Cryptographically Sealed)	Core Structural Layout	6+1 Node Molecular Matrix

↑ SYSTEM OPERATIONAL WARNING & MANDATORY BOUNDARIES

This document serves as the absolute, unified standalone specification for the PhytoIntelligence 3.4/3.5 system architecture. All prior documentation, including versions 3.3, 3.4, and intermediate amendment logs, are officially superseded. The formulations, multi-pass logic algorithms, and screening parameters described herein represent hypothetical deconvolution models derived from digitized peer-reviewed literature. No clinical validations have been executed. Do not use for human administration or real-world compound deployment without formal bench-scale validation and qualified medical oversight.

I. CORE ARCHITECTURAL & EPISTEMIC PRINCIPLES

The PhytoIntelligence (PI) framework operates under a zero-trust computational environment regarding biological data acquisition. Traditional models suffer from severe literature-drift, linear synergy assumptions, and baseline hallucination loops introduced by language-model synthesis layers. PI 3.5 enforces hard computational constraints to isolate multi-target natural compound architectures against specific molecular pathways.

1.1 The 1=1=1 Epistemic Axiom (Seshat's Axiom)

The system treats truth data as an absolute identity loop: an observation equals its replication, which equals its quantitative molecular realization (*Observation = Replication = Realization*). If a publication claims a phytochemical demonstrates an effect, that record is locked out unless it can satisfy the strict mathematical requirements of the Landry Hallucination-Free Protocol (LHFP-II).

1.2 Cryptographic Citation Hashing Protocol (CCHP)

To eliminate metadata contamination, citation drift, or retrospective data manipulation, all literature evidence feeding the underlying vector engine must be cryptographically anchored. When a computation stack is initiated, the vector engine compiles a string array of all references (PMIDs and DOIs):

```
String Array = [DOI 1, PMID 1, DOI 2, PMID 2 ...]
```

This array is passed through a SHA-256 cryptographic pipeline, outputting a unique, permanent Epistemic Fingerprint Hash which is permanently appended to the final formulation matrix. Any subsequent modification of the data layer automatically invalidates the stack signature.

1.3 Retraction Audit Trails & Cascade Downgrades

If a peer-reviewed study used to construct a node configuration is retracted, corrected, or formally questioned in international indices:

- The source study is instantly assigned an absolute retraction vector ($R_v = 0.0$).
- The affected compound's specific data points are forcefully downgraded to Tier 4 (QSAR modeling) by default.
- An immediate, automated recalculation loop triggers across the active framework repository. If the compound's localized Composite Evidence Score drops by more than 25% ($ACES > -25\%$), the system permanently invalidates the stack and halts the formulation compiler.

II. THE 12-STEP OPERATIONAL SEQUENCE

PI 3.5 expands the core formulation logic into a strict, sequentially dependent 12-step pipeline. No step may be bypassed, deferred, or calculated asynchronously.

Step	Protocol Name	Operational Scope & Execution Requirements	Gate/Threshold Criteria
01	Anomaly Observation	Identify the targeted pathological state or clear disease diagnosis. Map all terminal clinical endpoints and document documented co-morbidities.	Manual Input Lock
02	Intel Requirement	Map the primary and secondary enzymatic, genomic, or metabolic targets driving the pathology via deep digitized vector parsing.	MiniPASS: ≥3 peer-reviewed citations per target
02a	Confidence Tier Pre-Screen	Filter all candidate phytochemicals through the strict 4-Tier hierarchy. Strip out speculative compound architectures immediately.	Hard Fail on Tier 4-only compounds
03	Pathological Hypothesis	Deconstruct the primary disease state drivers into the 6-Dimensional Diagnostic State Vector (DSV). Assign components to specific nodes.	Formal DSV Alignment Matrix output
04	Vector Convergence Mapping	Execute an extraction pass of all peer-reviewed extraction data to establish raw docking potentials, binding affinities, and target matches.	Receptor matching score ≥ 0.75
05	Deconvolution Scoring	Compute the raw Composite Evidence Score (CES) for each candidate compound across all mapped pathways.	CES calculation baseline initialization
06	Node Assignment		DMCF Max Allocation Rule

		Populate the 6+1 Molecular Stack. Execute the new 3.5 Dynamic Matrix Contribution Factor loop to handle edge-case assignment ties.	
07	Pharmacokinetic Filter	Run the Tier-Weighted Interspecies Extrapolation model to compute the true Pharmacokinetic Alignment Score (PKAS).	PKAS(i) ≥ 0.50 for all core nodes
08	Mass Allocation (CWMA)	Distribute total targeted capsule mass (750mg baseline) using the iterative CES-Weighted Mass Allocation algorithm with a 5mg limit.	Max 5 iterations; hard 5mg floor protection
08a	Excipient Compatibility	Calculate the Hydrophobic Adsorption Potential (HAP) of the stack against the intended microcrystalline carrier or vehicle matrix.	HAP flag requires alternative non-adsorbing matrix
09	Synergy Verification	Construct the 6x6 Network Synergy Matrix. Apply the new 3.5 absolute double-pass filter to block internal antagonistic masking.	Pass 1: min(SI) ≥ 0.80 ; Pass 2: NSI ≥ 0.90
10	Safety Interaction Gate	Compute the global Safety Interaction Index (SII), integrating population polypharmacy data and checking Node 7 metrics.	SII ≤ 0.25 (Low/Negligible Risk Envelope)

III. THE 6-DIMENSIONAL DIAGNOSTIC STATE VECTOR (DSV)

The Diagnostic State Vector (DSV) is the definitive analytical core of the framework. Each candidate molecule is scored across six completely orthogonal biological dimensions. The resulting vector dictates structural node assignment and weight allocation.

Dimension	Scope & Pathway Engagement	Scoring Rubric Matrix (0 to 10 Points)	Target Node Placement
D1: Genomic	Direct modulation of gene expression, transcription factor activity, and oncogene/tumor-suppressor tracking (e.g., mTOR, NF-κB).	0 = No genomic trace; 5 = In-vitro transcriptomic evidence; 10 = Replicated clinical RCT confirmation.	Node 1 (Primary Driver)
D2: Metabolic Flux	Modulation of glycolysis, OXPHOS, lipid metabolism, fatty acid oxidation, and transmembrane transport mechanics.	0 = No trace; 5 = Cell-line mass-spectrometry metabolomics; 10 = In-vivo metabolic flux tracking.	Node 2 (Secondary Driver)
D3: Redox	Free radical scavenging, Nrf2/HO-1 nuclear translocation activation, and lipid peroxidation inhibition pathways.	0 = No evidence; 5 = Isolation assays (DPPH/FRAP); 10 = In-vivo confirmed redox homeostasis modification.	Node 3 (Synergy Node A)
D4: Immune	Cytokine signaling cascade modulation (IL-23/IL-17, TNF-α, IL-6), macrophage	0 = Unmapped; 5 = Clear in-vitro cytokine suppression; 10 =	Node 4 (Synergy Node B)

	polarization, and inflammasome inhibition.	Confirmed in-vivo immune contexture remodeling.	
D5: Epigenetic	Histone deacetylase (HDAC) inhibition, DNA methyltransferase modulation, and non-coding miRNA expression adjustments.	0 = No evidence; 5 = In-vitro epigenetic marker shifts; 10 = Human clinical epigenetic profiling validation.	Node 5 (Epigenetic Modifier)
D6: Kinetic	Endogenous clearance, hepatic first-pass stability, blood-brain barrier penetration, and phase-II conjugation escape profiles.	0 = Untracked; 5 = In-vitro microsomal half-life mapping; 10 = Human pharmacokinetic baseline tracking.	Node 6 (Kinetic Adjuvant)

3.1 Node-Assignment Tie-Break Engine

If two candidate compounds register identical CES scores for a specific dimensional peak, version 3.4 creates systemic calculation deadlocks. PI 3.5 forces a resolution by executing a Dynamic Matrix Contribution Factor (DMCF) loop. Let c_1 and c_2 be the tied entities; the engine evaluates their compatibility across all other active positions in the stack:

$$DMCF(C_x) = \sum_{j=1}^{N_{active}} (SI(C_x, N_j) \times OBC(C_x))$$

The entity matching the higher absolute DMCF is locked into the primary node slot, while the secondary entity is automatically routed to the auxiliary screening matrix.

IV. ADVANCED MATHEMATICAL FORMULATIONS & MULTI-PASS GATES

4.1 The Calibrated Bounded Risk Penalty Factor $\rho(R, TCI)$

Version 3.4 introduced a bounded risk component to patch mathematical singularities where risk and cost dropped beneath unity. However, the scalar values remained uncalibrated. PI 3.5 introduces calibrated scalars optimized against a matrix of known botanical toxicological records:

$$\rho = 1 + \alpha \ln(1 + \beta \cdot R \times TCI)$$

Where R is the literature-derived Adverse Event Risk Score ($0.0 \leq R \leq 10.0$) and TCI is the Translational Cost Index — a log-scaled human-equivalent dose burden proxy mapping target milligram ranges to toxicity profiles ($0.0 \leq TCI \leq 5.0$). The constants are structurally locked to $\alpha = 0.4343$ (Base-10 logarithmic alignment scale) and $\beta = 1.25$ (Toxicity Volatility Scale). This calibration ensures that severe toxic expressions or extreme dose demands trigger an exponential penalty curve that instantly deconstructs the compound's final Composite Evidence Score (CES).

4.2 The Network Synergy Index (NSI) Multi-Pass Filter

A critical vulnerability in version 3.4 allowed a single highly antagonistic compound pair (e.g., an antagonistic interaction showing an $SI_{ij} = 0.55$) to slip through the system undetected because a series of high-performing

synergy nodes inflated the geometric mean above the aggregate ≥ 0.90 threshold. PI 3.5 implements a mandatory Two-Pass Hard-Floor Gate:

- **Pass 1: Pairwise Boundary Evaluation.** The complete 6×6 matrix of all 15 pairwise interactions is checked. If any single pair fails to meet the hard floor, an immediate Node Disqualification occurs:

$$\min(SI_{ij}) \geq 0.80$$

- **Pass 2: Aggregate Network Evaluation.** Only if Pass 1 returns a clean true matrix is the global geometric mean computed and checked against the baseline standard:

$$NSI = (\prod_{k=1}^{15} SI_k)^{1/15} \geq 0.90$$

4.3 CES-Weighted Mass Allocation (CWMA) Convergence Limits

When distributing mass inside the physical 750mg target capsule shell, individual nodes registering a calculated mass of $m(i) < 5mg$ must be brought to a hard floor to remain physically uniform during manufacturing. In PI 3.4, redistributing this shortfall frequently triggered secondary floor violations in adjacent low-scoring nodes, causing infinite calculation loops. PI 3.5 implements a hard pythonic loop parameter to guarantee calculation convergence:

```
def cwma_allocation_engine(initial_masses, floor=5.0, max_iterations=5):
    iteration = 0
    masses = initial_masses.copy()
    while iteration < max_iterations:
        under_floor = [i for i, m in enumerate(masses) if m < floor and m > 0]
        if not under_floor:
            return masses # Convergence reached successfully

        shortfall = sum(floor - masses[i] for i in under_floor)
        for idx in under_floor:
            masses[idx] = floor

        eligible_nodes = [i for i, m in enumerate(masses) if m > floor]
        if not eligible_nodes:
            raise FrameworkDeadlockException("CWMA Pool Exhausted.")

        total_eligible_mass = sum(masses[i] for i in eligible_nodes)
        for e_idx in eligible_nodes:
            masses[e_idx] -= shortfall * (masses[e_idx] / total_eligible_mass)

        iteration += 1

    raise ConvergenceLimitException("CWMA Deadlock: Force Stack Redesign.")
```

V. PHARMACOKINETIC, SAFETY & EXCIPIENT OVERHAULS

5.1 Tier-Weighted PKAS Interspecies Extrapolation

Pharmacokinetic Alignment Scores (PKAS) assess whether different nodes will reach peak plasma concentrations (T_{max}) inside a unified window to allow synergy. PI 3.5 applies a strict penalty scalar to animal or cell-line derived pharmacokinetic parameters to eliminate false translational assumptions:

$$PKAS(i) = 1 - (|T_{max}(i) - T_{max}(Node\ 1)| / T_{max_window}) \times [1 + (1 - W_{tier})]$$

Where T_{max_window} is locked to a default value of 4 hours, and the Tier Discount Weight (W_{tier}) is selected strictly based on experimental source profiles:

- **Tier 1 (T1):** Human Clinical Trials / Validated In-Vivo Humans: $W_{tier} = 1.00$ (Zero Penalty)
- **Tier 2 (T2):** Non-Human Primate / Rodent In-Vivo Model: $W_{tier} = 0.70$ (30% Alignment Boundary Penalty)
- **Tier 3 (T3):** In-Vitro Caco-2 Cell Line Monolayer Permeability: $W_{tier} = 0.40$ (60% Alignment Boundary Penalty)
- **Tier 4 (T4):** In-Silico QSAR Prediction Engine: $W_{tier} = 0.00$ (Absolute Exclusion from PKAS Validation)

5.2 Safety Interaction Index (SII) Population-Sourcing & Node 7 Lockdown

The Safety Interaction Index (SII) screens a candidate stack for adverse interactions against conventional pharmaceutical regimens. PI 3.5 permanently introduces two critical modifications:

1. **Empirical Polypharmacy Integration:** The parameter *Concomitant_probability* (C_p) can no longer be estimated or guessed by developers. It must be derived programmatically from validated population-level prescription databases (such as CDC-NHANES or the FDA Sentinel System) mapped specifically to the disease demographic. For example, when running a psoriasis-targeted blueprint (B1.7), the system hard-locks prescription probability factors for co-administered drugs: $C_p(\text{Statins}) = 0.342$; $C_p(\text{NSAIDs}) = 0.512$. If localized demographic tracking data is missing, the system injects a conservative safety default of $C_p = 0.50$ automatically.
2. **Universal Node 7 Summation:** Any structural bioavailability enhancer, metabolic blocker, or P-glycoprotein efflux pump inhibitor assigned as an auxiliary agent (Node 7, e.g., Piperine, Naringin) must be explicitly summed within the primary calculation loop:

$$SII = [\sum_{i=1}^7 (CYP_inhibition(i) \times ADI_severity(i) \times C_p(i))] / N$$

Because Node 7 chemical architectures function specifically by down-regulating clear metabolic clearance pathways, their individual *CYP_inhibition* parameter is locked by default at a value of ≥ 0.85 . This configuration forces the overall stack to present near-zero baseline interactions across Nodes 1–6 to avoid violating the absolute safety threshold ($SII \leq 0.25$).

5.3 Excipient-Compound Adsorption Screening (Step 08a)

PI 3.5 integrates a mandatory physical carrier screening check to verify that microcrystalline cellulose nanosheet matrices (USP-MCC) do not compromise formulation dissolution via localized structural trapping. The system calculates the Hydrophobic Adsorption Potential (HAP) for all inputs:

$$HAP\ Trigger\ Condition = [\log P > 4.5]\ AND\ [PSA < 60\ \text{\AA}^2]$$

If any single phytochemical matches this structural threshold, the system flags the vehicle profile. The USP-MCC carrier must be substituted with an alternative non-adsorbing matrix (e.g., USP Dicalcium Phosphate) and documented in the final report.

VI. QUALITY CONTROL & REGULATORY SPLIT-PATHING

6.1 Inter-Batch Standardization Index ($S_I I$)

To correct for natural variance in wild botanical inputs, unpurified extracts must feature a standardized passport specification:

$$S_I I = (\% \text{ Target Active Marker Compound} / \text{Total Extract Solid Mass}) \times \theta_{\text{stability}}$$

Where $\theta_{\text{stability}}$ represents the verified shelf-decay coefficient across a 365-day environmental chamber cycle.

6.2 Regulatory Bifurcated Development Pipelines

To resolve deep vocabulary conflicts within the previous framework titles, PI 3.5 forces a rigid divergence in development pathways based upon the targeted regulatory filing:

Specification Parameter	Pathway A: Dietary Supplement	Pathway B: Investigational Botanical Drug (IND)
Framework Nomenclature	Phytochemical Synergy Layout (PSL)	Forensic Pharmacology Specification (FPS)
Target Envelope	Functional Health / Structure-Function Profile	Deep Pathological Intervention / Disease Eradication
Evidence Threshold	Tiers 1–3 Permitted; Zero Treatment Claims	Tier 1 (Human RCT) Mandatory for Core Primary Drivers
Carrier Matrix	USP-Grade Clean Label (Rice Hull, Cellulose)	Targeted Nanoparticles / Specialized Polymeric Delivery
Maximum Allowed SII	$SII \leq 0.10$ (Absolute Negligible Risk Floor)	$SII \leq 0.50$ (Monitored / Clinically Controlled Environment)

VII. APPENDIX: MODEL VALIDATION VERIFICATION DATASET

To allow external developers to check their codebases against the reference engine architecture of version 3.5.0, the following three-node trial simulation is provided.

Input Parameters

- **Node 1 (Primary):** $CES = 8.50$, $T_{\text{max}} = 1.5h$, Data Sourcing = Tier 1 Human RCT
- **Node 2 (Secondary):** $CES = 8.50$, $T_{\text{max}} = 3.5h$, Data Sourcing = Tier 2 Rodent In-Vivo Model ($w_{\text{tier}} = 0.70$)
- **Inter-Node Synergy Matrix Values:** $SI_{12} = 0.82$, $SI_{13} = 0.95$, $SI_{23} = 0.58$
- **Target Window Constraint:** $T_{\text{max_window}} = 4.0h$

Reference System Processing Steps & Expected Outputs

1. **Tie-Break Trigger Loop:** Because Node 1 and Node 2 share an identical Composite Evidence Score (8.50), the engine halts standard loading and executes the Section 3.1 DMCF matrix calculation loop. Node 1 secures the primary slot due to superior Tier 1 pharmacokinetic alignment values.

2. **PKAS Calculation:** The system evaluates Node 2's alignment against the Primary Driver:

$$PKAS(2) = 1 - (|3.5 - 1.5| / 4.0) \times [1 + (1 - 0.70)] = 1 - (2.0 / 4.0) \times 1.30 = 1 - 0.65 = 0.35$$

CRITICAL WARNING: $PKAS(2) = 0.35$ falls below the standard threshold of ≥ 0.50 . Node 2 is flagged as requiring specialized enteric-coating isolation or an independent dosing schedule.

3. **Network Synergy Index (NSI) Multi-Pass Filter:**

Pass 1 (Floor Verification): The engine scans the array of pairwise scores: [0.82, 0.95, 0.58]. It identifies that $\min(SI) = SI_{23} = 0.58$.

System Response: IMMEDIATE REJECTION. The score 0.58 drops beneath the hard floor barrier of 0.80. The stack is completely disqualified at Pass 1. The compiler refuses to evaluate the overall geometric mean, blocking an unsafe or highly antagonistic compound configuration from slipping through on an inflated average.

EPISTEMIC SECURITY SEAL CRYPTOGRAPHIC VERIFICATION MANIFEST

SHA-256: e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855

PI-v3.5.0-STANDALONE-SECURE // SYSTEM AUTHORIZED: M.S.S.L.